# RISHI KIRAN KARNATAKAM

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# EDUCATION

**NNRESGI** Bachelor of Technology, Computer Science and Engineering
Dec 2021 – Present | GPA: 8.4

**KMIIT** Intermediate (10+2 equivalent)
Sep 2019 – Jun 2021 | GPA: 9.4

**SAGHS** Secondary School Certificate
May 2019 | GPA: 9.8

# SKILLS

**Programming Languages:** C, Python
**Databases:** MySQL, MongoDB
**Cloud/DevOps:** Amazon Web Services, Git, Github, Jenkins
**Frameworks/Libraries:** TensorFlow, Flask

# CERTIFICATIONS

**SUPERVISED MACHINE LEARNING: REGRESSION AND CLASSIFICATION** Stanford University (Coursera)
Jun 2023

**THE JOY OF COMPUTING USING PYTHON** IIT Madras (NPTEL)
Jul 2023

**PYTHON FOR DATA SCIENCE** IIT Madras (NPTEL)
Jan 2025

**AICTE VIRTUAL INTERNSHIP** Nalla Narasimha Reddy Education Society's Group of Institutions
2024

**AWS ACADEMY GRADUATE - AWS ACADEMY CLOUD ARCHITECTING** AWS Academy
Sep 2024

# PROJECTS

**COTTON PLANT DISEASE DETECTION AND CLASSIFICATION USING TRANSFER LEARNING** Python, TensorFlow
Feb 2024

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- Developed a machine learning model to classify various diseases on cotton leaves through image processing techniques.
- Improved disease detection accuracy by **25%** using transfer learning with pre-trained CNN models.
- Achieved **90%+** accuracy on test datasets, optimizing performance for agricultural disease diagnosis.

**AI-POWERED CHESS ENGINE WITH GENETIC ALGORITHM OPTIMIZATION** Python, Tkinter, Chess Library
Aug 2024

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- Developed an AI-driven chess engine utilizing minimax, alpha-beta pruning, and Genetic Algorithms for optimized decision-making.
- Enhanced strategic depth and move accuracy by **20%**.
- Achieved a **30%** improvement in move generation speed through alpha-beta pruning.

**AI-INTEGRATED PLANT DISEASE DETECTION MOBILE APP** Flutter, Dart, TensorFlow Lite, MongoDB Atlas
Apr 2025

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- Built a mobile app for real-time plant disease detection using camera or gallery images.
- Integrated a **TensorFlow Lite** model (Inception V3) for accurate disease prediction with offline support.
- Enabled local data storage and automatic syncing to **MongoDB Atlas** when connected online.
- Designed a clean, modern UI with a sidebar to view history, profile, and AI predictions.

**INTERACTIVE SOLAR SYSTEM MODEL AND 3D GAME DEVELOPMENT** C#, Unity Engine
Oct 2023

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- Developed a **3D interactive solar system model** to enhance gamified learning.
- Created a first-person **3D game** inspired by Flappy Bird, leveraging dynamic renderings and immersive gameplay elements.

# AWARDS

**INTERNAL HACKATHON ON GENERATIVE AI** Achieved 3rd place in a college-hosted Hackathon focused on Generative AI.
Aug 2024

**NATIONAL HACKATHON ON AR/VR DEVELOPMENT** Ranked in the top 10 at a national-level hackathon focused on AR/VR.
Oct 2023

**NATIONAL HACKATHON ON GENERATIVE AI** Achieved 2nd place in a National Level Hackathon focused on Generative AI